Junseop Byun

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PROFILE

Computer Science student focusing on systems programming and network infrastructure. Have gained practical understanding by contributing to DPU-accelerated service mesh research and building a custom x86-64 compiler. Motivated by the challenge of solving low-level hardware-software bottlenecks and further expanding my knowledge in distributed systems.

EDUCATION

Seoul National University (SNU)

B.S. in Computer Science and Engineering

March 2022 - August 2027 (Expected)

- **GPA:** 3.95/4.30
- **Relevant Coursework:** Computer Network, System Programming, Computer Architecture, Quantum Computing, Theoretical Physics
- **Interests:** High-Performance Networking, System Programming

Seoul Science High School (SSHS)

March 2019 - February 2022

- **GPA:** 4.18/4.30

HONORS & AWARDS

- Second prize - Graduation Research Paper Presentation Contest (Physics, SSHS) (2021)
- Participant - 33rd Final Korean Mathematical Olympiad (FKMO) (2019)

SKILLS

- **Languages:** C/C++, Python, Java, OCaml, Go, SQL
- **Systems & Networking:** Linux System Programming, k8s, Service Mesh (Istio, Envoy, Linkerd), DPU (BlueField-3), DMA
- **Tools & Frameworks:** DOCA, Git, Qiskit, Django, Jetpack Compose

EXPERIENCE

Research Intern | *TNET lab* | *DPUmesh Project* | December 2025 – Present

- **DPU-Accelerated Service Mesh:**
  - Researching and optimizing service mesh architectures using Data Processing Units (DPU) to reduce CPU overhead.
  - Linkerd Offload: Running the Linkerd data plane directly on the BlueField-3 DPU, so sidecar proxying executes off the host CPU.
  - Transparent Framework Integration: Integrated the DPUmesh architecture directly into the gRPC framework, enabling existing applications to leverage the new service mesh with zero modifications to user code.
- **Sidecar Benchmarking & System Debugging:**
  - Benchmarked proxy overhead via DeathStarBench and resolved severe Consul registry leaks causing CPU and connection stalls.

PROJECTS

SNUPL2 Compiler | *Developer* | September 2023 - December 2023

- Developed a full-stack compiler for SNUPL/2, a custom educational programming language, completely from scratch using C.
- Managed the end-to-end compilation pipeline, including lexical analysis, parsing, Abstract Syntax Tree (AST) construction, and semantic analysis.
- Implemented the compiler backend to generate x86-64 assembly code, directly handling low-level register allocation.

MomenTag | *Full-Stack Developer* | September 2025 – December 2025

- Built a tag-based photo management application from scratch.
- Designed and integrated a RESTful backend using Django with a modern Android frontend using Jetpack Compose.

ADDITIONAL INTERESTS

- **Languages:** Korean, English.
- **Physics Studies:** Studied Classical Mechanics, Electrodynamics, and Quantum Information. Completed graduate-level coursework in Quantum Mechanics.